Smartshop v3

Software Design Document (SDD)

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Framework:	React (Vite) + Vercel Node Serverless + Supabase (Postgres)
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1. Executive Summary & Purpose

This Software Design Document (SDD) outlines the logical, physical, and database-level architecture of Smartshop v3. Smartshop is an industry-grade, highly conversion-optimized, mobile-responsive Telegram Mini App (TMA). It serves as a single codebase delivering state-of-the-art e-commerce features (Group Buying, Driver Express Delivery, and Vendor Subscriptions) to users across East Africa. By utilizing Vercel's serverless edge and Supabase's scalable PostgreSQL infrastructure, the system enforces strict transactional consistency and lightning-fast edge response times.

1.1 Scope & Boundaries

The application integrates three distinct user roles directly into a single unified client: Customer, Vendor, and Driver. This document details the critical backend routes inside `api/index.ts`, custom frontend React state bindings, and database schemas supporting features like dynamic geolocation detection, custom time limit restrictions, and inline secure driver PIN verification.

2. Detailed Module Architecture

To ensure a seamless, high-fidelity experience, the application segregates functions into isolated modules that communicate asynchronously via lightweight serverless REST API endpoints.

2.1 AI Voice-Note Ordering

The customer is presented with an integrated microphone recorder inside the main product search bar. Once recorded, the raw audio blob is sent serverlessly to the `/api/ai/voice-order` endpoint, which translates the voice to text and queries the database for matching products using SQL keyword matches, automatically adding the items to the shopping cart.

2.2 Traveling Salesman Route Optimizer

Drivers carrying multiple active deliveries are guided by an automated **Nearest-Neighbor Traveling Salesman (TSP)** routing algorithm. The Driver Dashboard calculates the distance vector between the driver's current position and all active dropoff coordinates, arranging their itinerary in the mathematically shortest, most fuel-efficient sequence.

2.3 Offline-Resilient Coordinates Cache

To protect courier tracking against cellular dead-zones, a silent background tracking thread caches coordinate updates in Local Storage whenever the internet connection drops, and auto-syncs the complete route retrospectively the second signal is restored.

3. Database Schema Design

The PostgreSQL schema inside Supabase uses strict relational integrity, indexed foreign keys, and custom Row Level Security (RLS) policies to guarantee multi-tenant safety and high-speed indexing.

Table Name	Primary Key	Key Columns	Indexes
subscription_plans	id (BIGINT)	name, category, daily_price, weekly_price, monthly_price, vendor_id, tags (JSONB)	idx_sp_category, idx_sp_active
subscriptions	id (BIGINT)	telegram_id, plan_id, status, price, next_delivery	idx_subs_plan, idx_subs_user, idx_subs_next
subscription_deliveries	id (BIGINT)	subscription_id, plan_id, status, delivery_date	idx_sd_subscription, idx_sd_date
deliveries	id (BIGINT)	order_number, driver_id, status, delivery_pin, customer_telegram_id	idx_del_status, idx_del_driver, idx_del_order

4. Core Algorithms (Pseudocode)

4.1 Nearest-Neighbor Traveling Salesman (TSP)

Below is the exact JavaScript algorithm used to optimize multi-stop driver delivery routes:

```
function optimizeRoute(driver_coords, active_deliveries) {
  let optimized = [];
  let unvisited = [...active_deliveries];
  let curr = driver_coords;

  while (unvisited.length > 0) {
    let closest_idx = 0;
    let min_dist = Infinity;
    for (let i = 0; i < unvisited.length; i++) {
      let dist = getDistance(curr, unvisited[i].coords);
      if (dist < min_dist) {
        min_dist = dist;
        closest_idx = i;
      }
    }
    let next_node = unvisited.splice(closest_idx, 1)[0];
    optimized.push(next_node);
    curr = next_node.coords;
  }
  return optimized;
}
```